Chapter 3

Advanced Derivative Topics

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The Constant e

By definition, $e = 2.718281828459\ldots$

e is also defined as the following limit:

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$$

Compound Interest

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$$

Compound interest: $FV = PV(1 + r/n)^{nt}$

Continuous compounding: $FV = PV\, e^{rt}$

r = annual interest rate

n = number of compound periods per year

t = time in years

Generous Grandma

Your Grandma puts $1,000 in a bank for you, at 5% interest. Calculate the future value after 20 years, if:

Compounded annually: $= 1000\ (1 + .05)^{20} = \$2,653.30$

Compounded daily

Assume 360 day count:

$$= 1000\left(1 + \frac{.05}{360}\right)^{(360)(20)} = \$2,718.09$$

Compounded continuously: $= 1000\ e^{(.05)(20)} = \$2,718.28$

$FV = PV\ e^{rt}$

IRA

$FV = PV e^{rt}$

After graduating from CU, you land a great job with a manufacturing firm. You contribute $3,000 to a Roth IRA and invest it in a mutual fund that grows at 12% a year, compounded continuously. If you plan to retire in 35 years, what will be its value at the end of the time period?

in excel, =exp(x) is $e^{(x)}$

Example

$FV = PV\, e^{rt}$

How long will it take an investment of \$5,000 to grow to \$8,000 if it is invested at 5% compounded continuously?

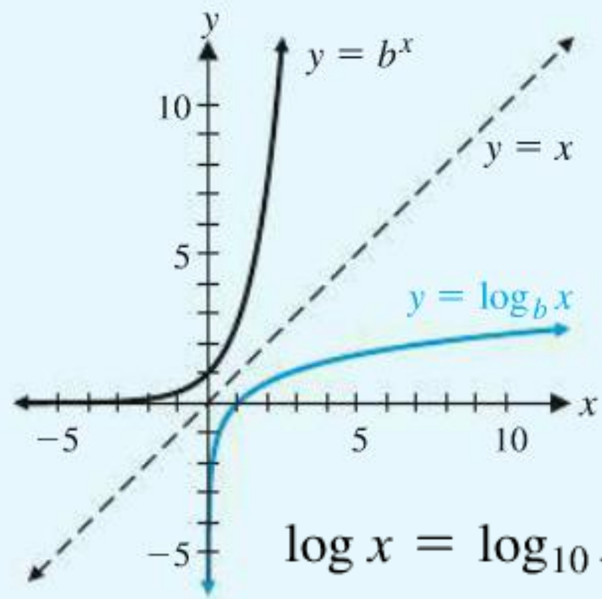
Logarithmic functions

SUMMARY

Recall that the inverse of an exponential function is called a **logarithmic function**. For $b > 0$ and $b \neq 1$,

Logarithmic form		Exponential form
$y = \log_b x$	is equivalent to	$x = b^y$

The graphs of $y = \log_b x$ and $y = b^x$ are symmetric with respect to the line $y = x$. (See Figure 1.)



$\log x = \log_{10} x$ Common logarithm (base 10)

$\ln x = \log_e x$ Natural logarithm (base *e*)

Logarithmic properties

THEOREM 1 Properties of Logarithmic Functions

If b, M, and N are positive real numbers; $b \neq 1$; and p and x are real numbers, then

1. $\log_b 1 = 0$
2. $\log_b b = 1$
3. $\log_b b^x = x$
4. $b^{\log_b x} = x, \quad x > 0$
5. $\log_b MN = \log_b M + \log_b N$
6. $\log_b \dfrac{M}{N} = \log_b M - \log_b N$
7. $\log_b M^p = p \log_b M$
8. $\log_b M = \log_b N$ if and only if $M = N$

Ln Returns (time additive)

$$\log_b M + \log_b N = \log_b MN$$

M	N	O	P
Date	Adj Close	Ln return	
2/5/2016	2.91	-0.03710375	=LN(N4/N5)
2/4/2016	3.02	0.03710375	=LN(N5/N6)
2/3/2016	2.91		

Derivatives of Exponential and Logarithmic Functions

The Derivative of e^x

$$\frac{d}{dx} e^x = e^x$$

The derivative of $f(x) = e^x$ is $f'(x) = e^x$

Caution: The derivative of e^x is not $x e^{x-1}$

The power rule cannot be used to differentiate the exponential function. The power rule applies to exponential forms x^n, where the exponent is a constant and the base is a variable

Examples

Find derivatives for:

$f(x) = e^x/2$

$f(x) = 2e^x + x^2$

$f(x) = -7x^e - 2e^x + e^2$

The Derivative of *Ln x*

$$\frac{d}{dx}\ln x = \frac{1}{x}$$

Examples

Find derivatives for:

$f(x) = 5 \ln x$

$f(x) = x^2 + 3 \ln x$

$f(x) = 10 - \ln x$

Derivatives

Chapter 3: total of 5 rules:

x^n (power)

e^x

$\ln(x)$

product rule

quotient rule

CHAIN RULE

Section 3.4

we MUST always completley differnciate a complex function

Derivatives of Exponential and Logarithmic Functions

For $b > 0$, $b \neq 1$,

$$\frac{d}{dx} e^x = e^x \qquad \frac{d}{dx} b^x = b^x \ln b$$

For $b > 0$, $b \neq 1$, and $x > 0$,

$$\frac{d}{dx} \ln x = \frac{1}{x} \qquad \frac{d}{dx} \log_b x = \frac{1}{\ln b}\left(\frac{1}{x}\right)$$

***See mathematical proofs (p.191 text) if desired**

Finance Example

$FV = PV e^{rt}$

An investment of $13,000 earns interest at an annual rate of 9%, compounded continuously.

" Derivative when $t = 4$ "

Find the instantaneous rate of change of the amount in the account after 4 years.

$FV(t) = 13{,}000\, e^{0.09t} \longrightarrow fv'(t) = 13000 \quad e^{0.09t}$

$FV'(4) = ?$

$Fv'(t) = 13000(0.09)e^{0.09t}$

Chain rule:

if $y = e^u$

$y' = e^u * u'$

$FV'(4) = (13000)(0.09)\, e^{(0.09 \cdot 4)}$

$= \exp(0.09 \cdot 4)$

→ EXCEL

r is the rate of interest and *t* is the time in years.

If $y = e^u$

$y' = e^u * u'$

$FV = PV\,e^{rt}$

Finance Example

PV

An investment of $13,000 earns interest at an annual rate of 9%, compounded continuously.

=rate

Find the instantaneous rate of change of the amount in the account when the account has a value of $14,000 = FV

$FV = PV\,e^{rt}$

$14{,}000 = 13{,}000\,e^{0.09t}$

$FV'(t) = ?$

$\frac{14000}{1300} = e^{0.09t}$

$\ln\left(\frac{14}{3}\right) = 0.09t$

$t = 0.8234$

$FV'(t) = \$1260$

$FV'(t) = PV \cdot r e^{rt}$

$r \cdot PV e^{rt} =$

they are the same

$FV'(t) = (r)(fv)$

If $y = e^{u}$

$y' = e^{u} * u'$

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$FV = PV\, e^{rt}$

Finance Example

An investment of \$13,000 earns interest at an annual rate of 9%, compounded continuously.

Find the instantaneous rate of change of the amount in the account when the account has a value of \$14,000

$14{,}000 = 13{,}000\, e^{0.09t}$

Or

$FV = PV\, e^{rt}$

$FV'(t) = r^* \, PV\, e^{rt}$

$FV'(t) = r^* \, FV$

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If $y = e^{u}$

$y' = e^{u} * u'$

Derivatives of Products and Quotients

$(a)(b)$

$$\frac{a}{b} \qquad \frac{9x-25}{x} = \frac{a}{b}$$

"use as desired"

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Derivatives of Products

THEOREM 1 Product Rule → implies one thing times another

F = "first"
S = "second"

F' = derivative
S' = derivative of second

If

$$y = f(x) = F(x)S(x)$$

and if $F'(x)$ and $S'(x)$ exist, then

$$f'(x) = F(x)S'(x) + S(x)F'(x)$$

The derivative of the product of two functions is:

the first function times the derivative of the second function plus the second function times the derivative of the first function

$$y' = FS' + SF'$$

$$y' = FS' + SF'$$

Example

Find the derivative of $y = (5x^2)(x^3 + 2)$

add exponets

$f'(x) = (5x^2)(3x^2) + (x^3+2)(10x)$

$15x$

$F = 5x^2$ $F' = 10x$

$S = (x^3+2)$ $S' = (x^3+2)$

$(5x^2)(3x^2) + (x^3+2)(10x)$

Solution:

$$f'(x) = 5x^2\,(3x^2) + 10x\,(x^3 + 2)$$

$$= 15x^4 + 10x^4 + 20x = \mathbf{25x^4 + 20x}$$

$$y' = FS' + SF'$$

Example

Find the derivative of $y = 2x^3(x^2 - 2)$

$F = 2x^3 \quad F' = 6x^2$

$S = (x^2-2) \quad S' = x^2$

$2x$

$2x^3(2x) + (x^2-2)\,6x^2$

$= 4x^4 + 6x^4 - 12x$

$= 10x^4 - 12x^2$

Solution:

$$f'(x) = 2x^3\,(2x) + (x^2 - 2)\,6x^2$$

$$= 4x^4 + 6x^4 - 12x^2 = \mathbf{10x^4 - 12x^2}$$

Derivatives of Quotients

order matters

THEOREM 2 Quotient Rule

*Start w/ bottom

watch your sign (SUBTRACTION)

If $\frac{T}{B} = \frac{\text{"top"}}{\text{"bottom"}}$

$$y = f(x) = \frac{T(x)}{B(x)}$$

and if $T'(x)$ and $B'(x)$ exist, then

$f'(x) = B(x)T'(x)$

$$f'(x) = \frac{B(x)T'(x) - T(x)B'(x)}{[B(x)]^2}$$

The derivative of the quotient of two functions is:

the bottom function times the derivative of the top function minus the top function times the derivative of the bottom function, all over the bottom function squared – ***order matters!***

Example

$$f'(x) = \frac{B(x)T'(x) - T(x)B'(x)}{[B(x)]^2}$$

Find the derivative of $y = 3x / (2x + 5)$

$B = (2x+5) \quad B' = 2$

$T = (3x) \quad T' = 3$

Solution:

$$\frac{(2x+5)(3) - (3x)(2)}{(2x+5)^2} = \frac{15}{(2x+5)^2}$$

$$f'(x) = \frac{D(x) \cdot N'(x) - N(x) \cdot D'(x)}{[D(x)]^2}$$

$$= \frac{(2x+5) \cdot 3 - 3x \cdot 2}{(2x+5)^2} = \frac{15}{(2x+5)^2}$$

Example

$$f'(x) = \frac{B(x)T'(x) - T(x)B'(x)}{[B(x)]^2}$$

$B = 2x+3$ $B' = 2$
$T = x^{-2}$ $T' = x$

$2x + 3x$ $x-$

Find the derivative of $y = (2x+3) / (x-2)$

Solution:

$\frac{-7x}{x^2}$

-4 -3

$$\frac{(x-2)(2) - ((2x+3)(1))}{(x-2)^2} = \frac{-7}{(x-2)^2}$$

$(x-2)(x-2)$

Video Game Sales

Total sales ***S*** (in thousands of video games) ***t*** months after the game is introduced is given by:

$$S(t) = \frac{125t^2}{t^2 + 100}$$

$B = 125t^2 = 250t$

$S = t^2 + 100 = S' = 2t$

$s(t^2+100)(250t) -$

B'

(S)

$$\frac{125(10)^2}{(10)^2+100} = 7812.5$$

$$\frac{250(10)}{2(t)} = \frac{2500}{20}$$

Find and interpret S(10) and S'(10)

Finally, estimate sales after 11 months

$S'(10) = 250$

$S(10) = 7812.5$

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Video Game Sales

$$S(t) = \frac{125t^2}{t^2 + 100}$$

$$f'(x) = \frac{B(x)T'(x) - T(x)B'(x)}{[B(x)]^2}$$

Find S(10), S'(10), and estimate sales after 11 months

$$S'(t) =$$

$$S(10) =$$

$$S'(10) =$$

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Video Game Sales

$$S(t) = \frac{125t^2}{t^2 + 100} \qquad S'(t) = \frac{25{,}000t}{(t^2+100)^2}$$

estimate sales after 11 months:

$S(10) =$ **62.5**

$S'(10) =$ **6.25**

SECTION 3.4

Chain Rule

the most missed questions on the exam!

$m(x) = f(g(x))$

Composite Functions

A function m is a composite of functions f and g if:

$$m(x) = f\,[g(x)]$$

General Power Rule

We have already made extensive use of the "BASIC power rule:"

$$\frac{d}{dx}x^n = nx^{n-1}$$

x^5

we want to generalize this rule so that we can differentiate composite functions of the form:

$[u(x)]^n$ where $u(x)$ is a differentiable function

$(x^2+2x+5)^3$

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Proof

Want
WBCH

Consider the derivative of $[u(\mathrm{x})]^2$

Product Rule:

$U(x) \cdot U(x)$

$f'(x) = U(x)U'(x) + U(x)U'(x)$

$f'(x) = 2\ u(x)u'(x)$

$f'(x) = 2u(x)$

$$\frac{d}{dx}[u(\mathrm{x})]^2 = \frac{d}{dx}[u(x)u(x)]$$

$$= u(x)u'(x) + (u(x)u'(x))$$

then multiply by derivative of the function

$$= \mathbf{2u(x)u'(x)}$$

THE Uncommitted closet

Share the ucommittment

Generalized Power Rule

THEOREM 1 General Power Rule

If $u(x)$ is a differentiable function, n is any real number, and

$$y = f(x) = [u(x)]^n$$

then

$$f'(x) = n[u(x)]^{n-1}\left(u'(x)\right) \longrightarrow$$

then multiply by derivative of the function

For example: $\dfrac{d}{dx}(x^2 + 3x + 5)^3 =$

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Example

$$\frac{d}{dx}u^n = nu^{n-1}\frac{du}{dx}$$

$$f(x) = (3x+1)^4$$

$$\boldsymbol{f'(x) = ?}$$

General Rules

General Derivative Rules

$$\frac{d}{dx}[f(x)]^n = n[f(x)]^{n-1}f'(x)$$

$$\frac{d}{dx}\ln[f(x)] = \frac{1}{f(x)}f'(x)$$

$$\frac{d}{dx}e^{f(x)} = e^{f(x)}f'(x)$$

If $y = u^n$, then
$y' = nu^{n-1}$ *u'

If $y = \ln u$, then
$y' = 1/u$ *u'

If $y = e^u$, then
$y' = e^u$ *u'

Find the Derivative

$$\frac{d}{dx}u^n = nu^{n-1}\frac{du}{dx}$$

$y = x^5,\ y' = 5x^4$

$y = (2x)^5$ $= 10x^4$

$y = (2x^3)^5$ $= 6x^2$

$y = (2x+1)^5$ $= 10x^4$

$y = (e^x)^5 =$ $5(e^x)^4$

$y = (\ln x)^5$ $= (5\ln x)^4$

Logarithmic Derivatives

$$\frac{d}{dx}\ln u = \frac{1}{u}\cdot\frac{du}{dx}$$

$y = \ln(4x)$, $= \frac{4}{x}$

$y = \ln(4x+1)$, $= \frac{4}{x+1}$

$y = \ln(x^2)$, $= \frac{1}{x}$

$y = \ln(x^2 + 2x - 4)$, $= 2x + x = 2x + \frac{1}{x}$

Exponential Derivatives

$$\frac{d}{dx}e^u = e^u \cdot \frac{du}{dx}$$

$$y = e^{3x} = 3e^{2x}$$

$$y = e^{3x+1} = 3e^{2x+1}$$

$$y = e^{4x^2-3x+5} \quad 4e^{3x^2-2x}$$

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Chain Rule

The **chain rule** can be extended to compositions of three or more functions. If y = f(w), w = f(u), and u = f(x), then:

$$\frac{dy}{dx} = \frac{dy}{dw}\frac{dw}{du}\frac{du}{dx}$$

Chain Rule

$$\frac{dy}{dx} = \frac{dy}{dw}\frac{dw}{du}\frac{du}{dx}$$

$$y = e^{1+(Ln\,x)^2}$$

$$\boldsymbol{\frac{dy}{dx} = ?}$$

Function Notation

So far, the equation of a curve has been specified in the form

$$y = x^2 - 5x \qquad \textbf{or} \qquad f(x) = x^2 - 5x$$

This is called the explicit form: y is given as a function of x

$2x - 5$

$2x - 5$

Explicit and Implicit Differentiation

explicit differentiation:

If $y = x^2 - 5x$

$\boldsymbol{y' = 2x - 5}$

implicit differentiation:

calculate the derivative of y from

$F(x, y) = x^2 - 5x - y = 0$

$2x - 5 - y$

Implicit Differentiation

$\mathbf{F(x, y) = x^2 – 5x – y = 0}$

$\boldsymbol{x^2 – y – 5x = 0}$

Differentiate both sides of the equation with respect to *x*, and keep in mind that *y* is a function of *x*

Example #1

Consider $x^2 - 3xy + 4y = 0$ and differentiate implicitly

$$\frac{d}{dx}x^2 - \frac{d}{dx}3xy + \frac{d}{dx}4y = \frac{d}{dx}0$$

Use the product rule for the 3*xy* term

Solve for y' :

Example #2

Consider $xe^x + \ln y + 3y = 0$ and differentiate implicitly

Solve for y':

Chain Rule !

CONCEPTUAL INSIGHT

When differentiating implicitly, the derivative of y^2 is $2yy'$, not just $2y$. This is because y represents a function of x, so the chain rule applies. Suppose, for example, that y represents the function $y = 5x + 4$. Then

$$(y^2)' = [(5x + 4)^2]' = 2(5x + 4) \cdot 5 = 2yy'$$

So, when differenting implicitly, the derivative of y is y', the derivative of y^2 is $2yy'$, the derivative of y^3 is $3y^2y'$, and so on.